

Food Waste Reduction and Donation Management System

Object Analysis & Analysis Classes

SE LAB 04 A

Submitted By:

Rudrakshi Chincholkar (241071014)

Nihira Neralwar (241071049)

April 5, 2026

Contents

1	Food Donor Use Cases	2
1.1	Register / Login (Food Donor)	2
1.2	Add/Edit/Cancel/View Food Request	2
1.3	Follow Food Safety Guidelines	2
1.4	Feedback & Rating System	2
2	NGO / Food Bank Use Cases	2
2.1	View Available Donations	2
2.2	Accept/Reject Food Donation	2
2.3	Manage Volunteers	2
2.4	Manage Delivery Partners	3
2.5	Penalize Food Donor	3
2.6	Assign / Manage Food Inspector	3
2.7	Manage Unsafe Food	3
2.8	Donation Matching System	3
3	Volunteer Use Cases	3
3.1	Report Issues in Pickup	3
3.2	Set Availability	3
3.3	View Pickup Requests	3
3.4	Accept Pickup Task	4
3.5	Update Pickup Status	4
4	Delivery Partner Use Cases	4
4.1	Accept Delivery Task	4
4.2	Pick Up Food	4
4.3	Deliver Food	4
4.4	Update Delivery Status	4
5	System Administrator Use Cases	4
5.1	Manage Users	4
5.2	Monitor System Activity	5
5.3	Generate Reports	5
5.4	Manage Records (Database)	5
5.5	Real-Time Notification System	5
5.6	Track Donation Lifecycle	5
6	Food Safety Inspector Use Cases	5
6.1	Inspect Food Donation	5
6.2	Define Food Safety Rules	5

1 Food Donor Use Cases

1.1 Register / Login (Food Donor)

Entity Objects: User, DonorProfile, Credentials

Boundary Objects: RegistrationFormUI, LoginFormUI

Control Objects: AuthenticationController, RegistrationController

1.2 Add/Edit/Cancel/View Food Request

Entity Objects: FoodDonation, FoodItem, Donor, Location, DonationStatus **Boundary Objects:** AddDonationFormUI, DonationListUI, EditDonationFormUI, DonationStatusDashboardUI

Control Objects: DonationController, DonationValidationController, DonationManagementController, DonationTrackingController

1.3 Follow Food Safety Guidelines

Entity Objects: FoodSafetyGuideline

Boundary Objects: GuidelinesViewUI

Control Objects: GuidelineController

1.4 Feedback & Rating System

Entity Objects: Feedback, Rating, FoodDonation, User

Boundary Objects: FeedbackFormUI, RatingUI

Control Objects: FeedbackController, RatingController

2 NGO / Food Bank Use Cases

2.1 View Available Donations

Entity Objects: FoodDonation, Donor, Location

Boundary Objects: AvailableDonationsUI

Control Objects: DonationController

2.2 Accept/Reject Food Donation

Entity Objects: FoodDonation, InspectionReport, NGO

Boundary Objects: DonationDecisionUI, InspectionFormUI

Control Objects: DonationController, InspectionController

2.3 Manage Volunteers

Entity Objects: PickupRequest, Schedule, Volunteer, DeliveryPartner

Boundary Objects: PickupSchedulingUI

Control Objects: PickupSchedulerController, TaskAssignmentController

2.4 Manage Delivery Partners

Entity Objects: DistributionRecord, FoodDonation, Beneficiary

Boundary Objects: DistributionManagementUI

Control Objects: DistributionController

2.5 Penalize Food Donor

Entity Objects: Donor, PenaltyRecord, Violation

Boundary Objects: DonorPenaltyUI

Control Objects: PenaltyController, ViolationAssessmentController

2.6 Assign / Manage Food Inspector

Entity Objects: FoodInspector, InspectionSchedule, Assignment

Boundary Objects: InspectorAssignmentUI

Control Objects: InspectorAssignmentController

2.7 Manage Unsafe Food

Entity Objects: FoodDonation, SafetyFlag, DisposalRecord

Boundary Objects: UnsafeFoodManagementUI

Control Objects: UnsafeFoodController, DisposalController

2.8 Donation Matching System

Entity Objects: FoodDonation, NGO, MatchingCriteria, Location

Boundary Objects: MatchingDashboardUI

Control Objects: MatchingController, RecommendationEngine

3 Volunteer Use Cases

3.1 Report Issues in Pickup

Entity Objects: IssueReport, PickupTask, Volunteer

Boundary Objects: IssueReportingUI

Control Objects: IssueController

3.2 Set Availability

Entity Objects: Volunteer, AvailabilitySchedule

Boundary Objects: AvailabilityFormUI

Control Objects: AvailabilityController

3.3 View Pickup Requests

Entity Objects: PickupRequest, Location

Boundary Objects: PickupRequestsListUI

Control Objects: PickupRequestController

3.4 Accept Pickup Task

Entity Objects: PickupTask, Volunteer, PickupRequest

Boundary Objects: TaskAcceptanceUI

Control Objects: PickupTaskController

3.5 Update Pickup Status

Entity Objects: PickupTask, PickupStatus

Boundary Objects: PickupStatusUpdateUI

Control Objects: PickupStatusController

4 Delivery Partner Use Cases

4.1 Accept Delivery Task

Entity Objects: DeliveryTask, DeliveryPartner

Boundary Objects: DeliveryTaskUI

Control Objects: DeliveryTaskController

4.2 Pick Up Food

Entity Objects: PickupTask, FoodDonation, Location

Boundary Objects: PickupConfirmationUI

Control Objects: PickupController

4.3 Deliver Food

Entity Objects: DeliveryTask, DeliveryStatus

Boundary Objects: DeliveryConfirmationUI

Control Objects: DeliveryController

4.4 Update Delivery Status

Entity Objects: DeliveryStatus, DeliveryTask

Boundary Objects: DeliveryStatusUpdateUI

Control Objects: DeliveryStatusController

5 System Administrator Use Cases

5.1 Manage Users

Entity Objects: User, Role, Permission

Boundary Objects: UserManagementDashboardUI

Control Objects: UserManagementController

5.2 Monitor System Activity

Entity Objects: ActivityLog, User

Boundary Objects: ActivityMonitoringUI

Control Objects: SystemMonitoringController

5.3 Generate Reports

Entity Objects: Report, DonationRecord, DistributionRecord

Boundary Objects: ReportGenerationUI

Control Objects: ReportController

5.4 Manage Records (Database)

Entity Objects: DatabaseRecord, BackupLog, AuditLog

Boundary Objects: DatabaseManagementUI

Control Objects: DatabaseController, BackupController

5.5 Real-Time Notification System

Entity Objects: Notification, User

Boundary Objects: NotificationUI

Control Objects: NotificationController, AlertDispatcher

5.6 Track Donation Lifecycle

Entity Objects: FoodDonation, TrackingLog, StatusHistory

Boundary Objects: DonationTrackingUI

Control Objects: TrackingController

6 Food Safety Inspector Use Cases

6.1 Inspect Food Donation

Entity Objects: InspectionReport, FoodDonation, FoodSafetyRule

Boundary Objects: InspectionFormUI

Control Objects: InspectionController

6.2 Define Food Safety Rules

Entity Objects: FoodSafetyRule

Boundary Objects: RuleDefinitionUI

Control Objects: FoodSafetyRuleController

6.3 Flag Unsafe Food

Entity Objects: FoodDonation, InspectionReport, SafetyFlag

Boundary Objects: UnsafeFoodFlagUI

Control Objects: SafetyFlagController